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Test Pattern

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

NEET(UG)

MINOR

05-02-2023

ENTHUSIAST ADVANCE COURSE PHASE : MEB-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

This Booklet contains 28 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 Hours 20 Minutes** duration and this Test Booklet contains **200** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. In this Test Paper, each subject will consist of **two sections**. **Section A** will consist of **35** questions (all questions are mandatory) and **Section B** will have **15** questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.
4. In case of more than one option correct in any question, the best correct option will be considered as answer.
5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

TG: @Chalnaayaaar

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2024

Topic : ELECTROSTATICS, PLANE MIRROR TO APPARENT DEPTH, UNIT AND DIMENSION

SECTION-A

Attempt All 35 questions

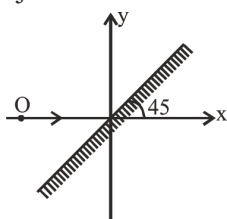
1. if an observer is walking away from the plane mirror with 6m/sec. Then the speed of the image with respect to observer will be :-

(1) 6 m/sec (2) -6 m/sec
(3) 12 m/sec (4) 3 m/sec

2. Find minimum length of mirror required for a person to see his complete image if person height is 5 feet.

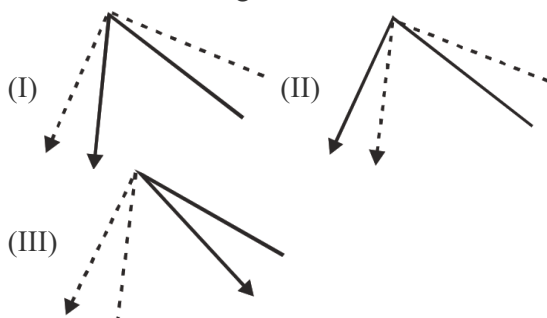
(1) 2 feet (2) 3 feet (3) 4 feet (4) 2.5 feet

3. Image of the object can be :-



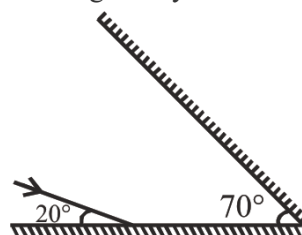
(1) ↓ (2) ↑ (3) → (4) ↖

4. Each of these diagrams is supposed to show two different rays being reflected from the same point on the same plane mirror. Which one of the following is correct ?



(1) Only I (2) Only II
(3) Only III (4) All

5. Deviation of emergent ray is :-



(1) 120 (2) 140 (3) 180 (4) 90

6. If a plane mirror is rotated by an angle 8° without changing the direction of incident ray then reflected ray is rotated by :-

(1) 2° (2) 4° (3) 8° (4) 16°

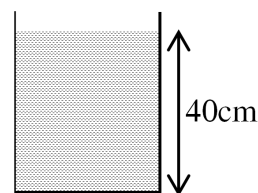
7. A convex mirror has a focal length f . A real object is placed at a distance f in front of it from the pole produces an image at

(1) Infinity (2) f (3) $f/2$ (4) $2f$

8. A square of side 3 cm is placed at a distance of 25 cm from a concave mirror of focal length 10 cm. The centre of the square is at the axis of the mirror and the plane is normal to the axis. The area enclosed by the image of the wire is -

(1) 4 cm^2 (2) 6 cm^2 (3) 16 cm^2 (4) 36 cm^2

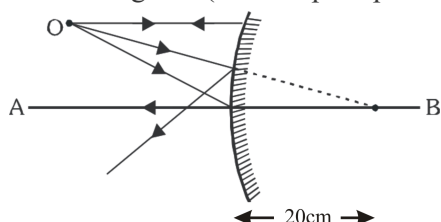
9. In the given figure an observer in air ($n = 1$) sees the bottom of a beaker filled with water ($n = 4/3$) upto a height of 40 cm. What will be the depth felt by this observer.



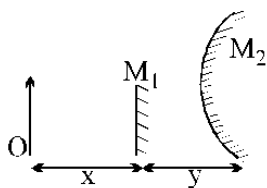
(1) 10 cm (2) 20 cm (3) 30 cm (4) 40 cm

ALLEN

10. The distance of an object from a spherical mirror is equal to the focal length of the mirror then the image :-
 (1) must be at infinity (2) may be at infinity
 (3) may be at the focus (4) none
11. In the figure a convex mirror of radius of curvature 20 cm is shown. An object O is placed in front of this mirror. Its ray diagram is shown. How many mistakes are there in the diagram? (AB is its principal axis) :-



- (1) 3 (2) 2 (3) 1 (4) 0
12. A particle is moving with velocity $(\hat{i} + 2\hat{j} + 3\hat{k})$ m/s. A plane mirror exists in the x-y plane. The mirror has a velocity $(\hat{i} + 2\hat{j})$ m/s. Velocity of the image (in m/s) is:-
 (1) $(\hat{i} + 2\hat{j})$ (2) $\hat{i} + 2\hat{j} - 3\hat{k}$
 (3) $-\hat{j} - \hat{k}$ (4) none of these
13. An object O is placed in front of a small plane mirror M_1 and a large convex mirror M_2 of focal length f . The distance between O and M_1 is x , and the distance between M_1 and M_2 is y . The images of O formed by M_1 and M_2 coincide. The magnitude of f is



- (1) $\frac{x^2 - y^2}{2y}$ (2) $\frac{x^2 + y^2}{2y}$
 (3) $x - y$ (4) $\frac{x^2 + y^2}{x - y}$

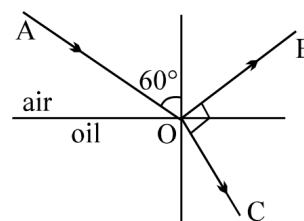
14. A bird in air looks a fish vertically below it and inside water ; h_1 is the height of the bird above the surface of water and h_2 the depth of the fish below the surface of water. If refractive index of water with respect to air be μ , then the distance of the fish as observed by the bird is :

- (1) $h_1 + h_2$ (2) $h_1 + \frac{h_2}{\mu}$
 (3) $\mu h_1 + h_2$ (4) $\mu h_1 + \mu h_2$

15. The refractive index of a certain glass is 1.5 for light whose wavelength in vacuum is $6000 \text{ \AA}$. The wavelength of this light when it passes through glass is

- (1) $4000 \text{ \AA}$ (2) $6000 \text{ \AA}$
 (3) $9000 \text{ \AA}$ (4) $15000 \text{ \AA}$

16. A ray of light AO is incident on the surface of oil. Reflected part of this ray OB and refracted part OC are mutually perpendicular as shown. Find refractive index of oil.



- (1) $\sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$
 (3) $\frac{1}{\sqrt{3}}$ (4) 2

17. A ray of light passes from glass having refractive index of 1.6 to air. The angle of incidence for which the angle of refraction is twice the angle of incidence is :-

- (1) $\sin^{-1}(4/5)$ (2) $\sin^{-1}(3/5)$
 (3) $\sin^{-1}(5/8)$ (4) $\sin^{-1}(2/5)$

ALLEN

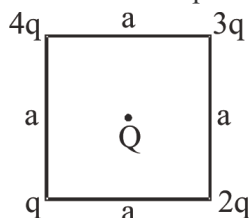
18. Electromagnetic energy is radiated by :-

- (1) Stationary charge
- (2) A charge in uniform motion
- (3) An accelerated charge only
- (4) An accelerated charge and a charge in uniform motion

19. A and B are two identical spherical charged bodies which repel each other with force F , kept at a finite distance apart. A third uncharged sphere of the same size is brought in contact with sphere B and removed. Find the magnitude of force on B :-

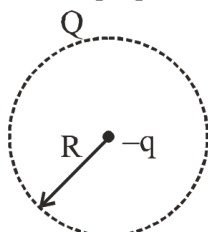
- (1) $F/2$
- (2) $F/8$
- (3) F
- (4) Zero

20. In figure, magnitude of force experienced by Q is :-



- (1) $\frac{1}{\pi\epsilon_0} \frac{qQ\sqrt{2}}{a^2}$
- (2) $\frac{1}{4\pi\epsilon_0} \frac{qQ}{a^2}$
- (3) $\frac{1}{4\pi\epsilon_0} \frac{qQ\sqrt{2}}{a^2}$
- (4) $\frac{1}{\pi\epsilon_0} \frac{qQ}{a^2}$

21. Charge Q of mass m revolves around a point charge $-q$ due to electrostatic attraction. its period of revolution is proportional to :-

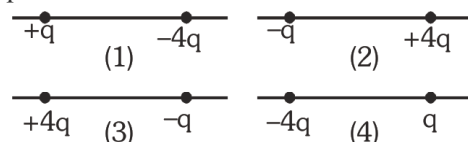


- (1) $Q^{1/2}$
- (2) $m^{1/2}$
- (3) R^3
- (4) q^2

22. In quark model, a proton and a neutron consist of three quarks each. Two types of quarks, the so called 'up' quark (denoted by u) of charge $+\frac{2}{3}e$ and the 'down' quark (denoted by d) of charge $-\frac{1}{3}e$, together with electrons build up ordinary matter. A possible mark composition of a proton and neutron is :-

- (1) udd, udd
- (2) uud, udd
- (3) uud, uud
- (4) ddd, uuu

23. The figure shows four situations in which charges as indicated ($q > 0$) are fixed on an axis. In which situation is there a point to the left of the charges where an electron would be in equilibrium ?



- (1) 1 and 2
- (2) 2 and 4
- (3) 3 and 4
- (4) 1 and 3

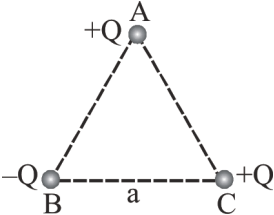
24. Mark correct option :-

- (1) The total charge of universe is constant
- (2) The total positive charge of universe is constant
- (3) The total negative charge of universe is constant
- (4) The total number of charge particles in universe is constant

25. When 10^{14} electrons are removed from a neutral metal sphere, the charge on it becomes:-

- (1) $16\mu C$
- (2) $-16\mu C$
- (3) $32\mu C$
- (4) $-32\mu C$

ALLEN

26. Two particles having charges Q_1 and Q_2 , when kept at a certain distance exert a force F on each other. If the distance between the two particles is reduced to half and the charge on each particle is doubled, the force between the particles would be :-
 (1) $2F$ (2) $4F$ (3) $8F$ (4) $16F$
27. An electron is moving round the nucleus of a hydrogen atom in a circular orbit of radius r . The Coulomb force $\vec{F}$ on the electron is :-
 (1) $K \frac{e^2}{r^2} \hat{r}$ (2) $-K \frac{e^2}{r^3} \hat{r}$
 (3) $K \frac{e^2}{r^3} \vec{r}$ (4) $-K \frac{e^2}{r^3} \vec{r}$
28. Consider a system of 3 bodies A, B and C having net charge 10 C. If charge on C is 2C then charges on A & B can be
 (1) $A = -4C, B = -4C$ (2) $A = 3C, B = 5C$
 (3) $A = -4C, B = 12C$ (4) Both 2 & 3
29. A body can be negatively charged by :-
 (1) Giving excess of electrons to it
 (2) Removing some electrons from it
 (3) Giving some protons to it
 (4) Removing some neutrons from it
30. Three charges are placed at the vertices of an equilateral triangle of side 'a' as shown in the following figure. The force experienced by the charge placed at the vertex A in a direction normal to BC is :-

 (1) $Q^2/(4\pi\epsilon_0 a^2)$ (2) $-Q^2/(4\pi\epsilon_0 a^2)$
 (3) Zero (4) $Q^2/(2\pi\epsilon_0 a^2)$
31. One full rotation of the cap of a screw gauge is equivalent to 5mm. The cap has 100 division. Find the least count:
 (1) 0.5 m (2) 0.05 mm
 (3) 0.005 m (4) 5×10^{-4} m
32. Which of the following has lowest number of significant figures ?
 (1) 4720 (2) 275000 m
 (3) 4.4000 (4) 0.002430
33. If P & Q have two different non-zero dimensions, which of the following operation is possible ?
 (1) $P + Q$ (2) PQ
 (3) $P - Q$ (4) $1 - \frac{P}{Q}$
34. The velocity of a particle (v) at an instant t is given by $v = at + bt^2$ the dimension of b is :-
 (1) L (2) LT^{-1} (3) LT^{-2} (4) LT^{-3}
35. In an experiment, four quantities a, b, c and d are measured with percentage errors 1%, 2%, 3% and 4% respectively. Quantity P is calculated as follows $P = \frac{a^3 b^2}{cd}$, then percentage error in P is :-
 (1) 4% (2) 14% (3) 10% (4) 7%

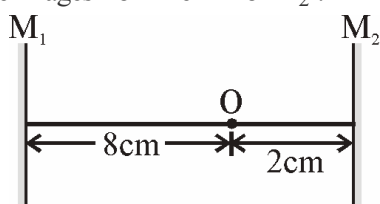
SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

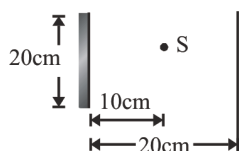
36. An object is at a distance of 0.5 m in front of a plane mirror. Distance between the object and image is :-
 (1) 0.5 m (2) 1 m (3) 0.25 m (4) 1.5 m

ALLEN

37. Figure below shows two plane mirrors and an object O placed between them. What will be distance of the first three images from the mirror M_2 ?

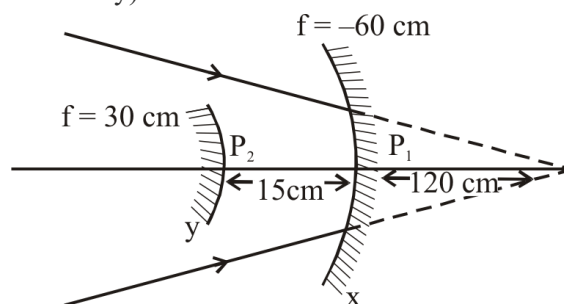


- (1) 2 cm, 8 cm, 14 cm (2) 2 cm, 12 cm, 18 cm
(3) 2 cm, 18 cm, 22 cm (4) 2 cm, 24 cm, 38 cm
38. The refractive indices of glass and water w.r.t air are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. The refractive index of glass w.r.t water will be
- (1) $\frac{8}{9}$ (2) $\frac{9}{8}$ (3) $\frac{7}{6}$ (4) None
39. A convex mirror has a radius of curvature 20 cm. If a point source is placed 14 cm away from the mirror, the image is formed
- (1) $\frac{70}{12}$ cm behind the mirror
(2) $\frac{70}{12}$ cm in front the mirror
(3) $\frac{80}{12}$ cm behind the mirror
(4) 5 cm in front the mirror
40. A point source of light S is placed at a distance 10 cm in front of the centre of a mirror of width 20 cm suspended vertically on a wall. A man walks with a speed 10 cm/s in front of the mirror along a line parallel to the mirror at a distance 20 cm from it as shown in figure. Find the maximum time during which he can see the image of the source S in the mirror :-



- (1) 6 s (2) 3 s (3) 2 s (4) 1 s

41. A fish in water sees an object which is 24 cm above the surface of water. The height of the object above the surface of water that will appear to the fish is ($\mu = \frac{4}{3}$)
- (1) 24 cm (2) 32 cm (3) 18 cm (4) 48 cm
42. A beam of light from a source is incident normally on a plane mirror fixed at a distance 2m from the source. The beam is reflected back as a spot on a scale placed just above the source. When the mirror is rotated through a small angle 1.8° , the spot of the light is found to move through a distance d on the scale. The distance d is :-
- (1) 0.1831 m (2) 0.1256 m
(3) 3.14 m (4) None of these
43. The image formed after two reflection (first at x then at y) will be :-



- (1) Erect and enlarged 2 times
(2) Inverted and enlarged 2 times
(3) Erect and diminished 2 times
(4) Inverted and diminished 4 times
44. Choose correct statement/s :-
- (A) In conduction charge transfer takes place till potential of both bodies becomes equal.
(B) Induced body will get opposite nature of charge.
(C) Induction is a particle phenomena.
- (1) Only A (2) Only B
(3) Both A and B (4) All

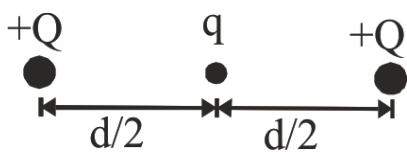
ALLEN

45. If linear charge density of a wire of length L depends on distance x from one end as $\lambda = \frac{\lambda_0(x)}{L}$.



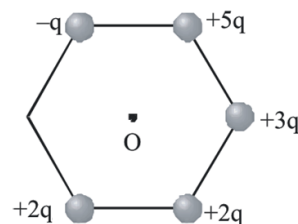
Find total charge of wire.

- (1) $\lambda_0 L$ (2) $\frac{\lambda_0 L}{2}$
 (3) $\frac{\lambda_0 L}{4}$ (4) None of these
46. Two particles A and B (B is right of A) having charges $8 \times 10^{-6} \text{ C}$ and $-2 \times 10^{-6} \text{ C}$, respectively, are held fixed with separation of 20 cm. Where should a third charged particle be placed so that it does not experience a net electric force :-
- (1) 5 cm right of B
 (2) 5 cm left of A
 (3) 20 cm left of A
 (4) 20 cm right of B
47. Two positively charged particles each having charge Q are at distance d apart. A third charge is introduced in midway on the line joining the two. Find nature and magnitude of third charge, so that the system is in equilibrium :

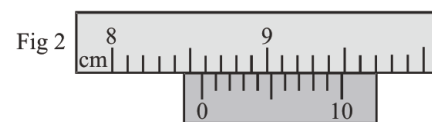
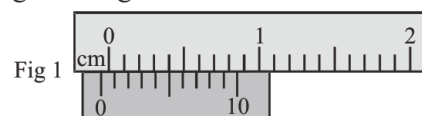


- (1) $q = \frac{-Q}{4}$ (2) $q = \frac{Q}{4}$
 (3) $q = \frac{3Q}{4}$ (4) $q = -\frac{3Q}{4}$

48. Five point charges are placed at the corners of a regular hexagon of side ' a ' as shown in figure. Net force experienced by an electron placed at centre O will be :-



- (1) $\frac{6kqe}{a^2}$ towards $+5q$ charge
 (2) $\frac{3kqe}{a^2}$ towards $+3q$ charge
 (3) $\frac{6kqe}{a^2}$ towards $+3q$ charge
 (4) $\frac{3kqe}{a^2}$ towards $-q$ charge
49. The main scale of a vernier caliper reads in mm and its Vernier is divided into 10 divisions which coincide with 9 divisions of the main scale. The reading for the given situation will be :-



- (1) 7.58 cm (2) 8.79 cm
 (3) 8.63 cm (4) 9.78 cm
50. The values of two resistance are $R_1 = (6 \pm 0.3) \text{ K}\Omega$ and $R_2 = (10 \pm 0.2) \text{ K}\Omega$. The % error in the equivalent resistance when they are connected in parallel is :-
- (1) 5.125% (2) 2%
 (3) 10.125% (4) 7%

Topic : PERIODIC TABLE (COMPLETE), CHEMICAL BONDING (TILL TAUGHT), CHEMICAL KINETICS, SOLUTIONS (TILL TAUGHT)**SECTION-A****Attempt All 35 questions**

51. First IE of 5d series elements are higher than those of 3d and 4d series elements. This is due to :

- (1) Bigger size of atoms of 5d-series elements than 3d-series elements.
- (2) Greater effective nuclear charge is experienced by valence electrons because of the weak shielding by 4f-electrons in 5d series.
- (3) (1) and (2) both.
- (4) None of these.

52. Out of N, O, Ne, Na, Ne^+ select the species having maximum & minimum I.P. respectively :-

- (1) Ne, Ne^+ (2) Ne^+ , Na
- (3) Ne, Na (4) Ne, N

53. Select the correct order of size for : Cl^- , O^{2-} , Na^+ , Br^- , Li^+

- (1) $\text{Br}^- > \text{O}^{2-} > \text{Cl}^- > \text{Na}^+ > \text{Li}^+$
- (2) $\text{Cl}^- > \text{Br}^- > \text{O}^{2-} > \text{Na}^+ > \text{Li}^+$
- (3) $\text{Br}^- > \text{Na}^+ > \text{Li}^+ > \text{O}^{2-} > \text{Na}^+$
- (4) $\text{Br}^- > \text{Cl}^- > \text{O}^{2-} > \text{Na}^+ > \text{Li}^+$

54. The element with strongest electropositive nature is :

- (1) Cu (2) Cs (3) Cr (4) Ba

55. Among the following configurations, the element which has the highest electron affinity is :-

- (1) $[\text{Ne}] 3s^2 3p^2$ (2) $[\text{Ne}] 3s^2 3p^5$
- (3) $[\text{Ne}] 3s^2 3p^4$ (4) $[\text{Ne}] 3s^2 3p^1$

56. Match the following

	Order		Property
(a)	$\text{Li} < \text{Be} < \text{B}$	(i)	Electron affinity
(b)	$\text{Be} < \text{B} < \text{Li}$	(ii)	Atomic radius
(c)	$\text{Li} < \text{B} < \text{Be}$	(iii)	Ionisation energy
(d)	$\text{Li} > \text{Be} > \text{B}$	(iv)	Z_{eff}

- (1) a – (iv); b – (i); c – (ii); d – (iii)
- (2) a – (i); b – (iv); c – (ii); d – (iii)
- (3) a – (iv); b – (i); c – (iii); d – (ii)
- (4) a – (iv); b – (iii); c – (i); d – (ii)

57. The last element of the p-block in the 6th period represented by the outermost electronic configuration

- (1) $7s^2 7p^6$ (2) $5f^4 6d^{10} 7s^2 7p^0$
- (3) $4f^{14} 5d^{10} 6s^2 6p^6$ (4) $4f^4 5d^{10} 6s^2 6p^4$

58. Magnetic moment of X^{3+} ion of 3d series is $\sqrt{35}$ BM. What is atomic number of X^{3+} ?

- (1) 25 (2) 26 (3) 27 (4) 28

59. The pair with minimum difference in electronegativity is :-

- (1) F, Cl (2) C, H (3) P, H (4) Na, Cs

60. Select equation having exothermic step

- (I) $\text{S}_{(\text{g})}^+ \rightarrow \text{S}_{(\text{g})}$ (II) $\text{N}_{(\text{g})}^+ \rightarrow \text{N}_{(\text{g})}$
- (III) $\text{N}_{(\text{g})} \rightarrow \text{N}_{(\text{g})}^-$ (IV) $\text{Al}_{(\text{g})}^{+2} \rightarrow \text{Al}_{(\text{g})}^{+3}$

Choose the correct code

- (1) II (2) I, II
- (3) III and IV (4) II and III

ALLEN

61. The correct order of Radius is -

- (1) $\text{Li}^+ < \text{Be}^{+2}$ (2) $\text{Mn} > \text{Sc}$
 (3) $\text{Li}^+ \simeq \text{Mg}^{+2}$ (4) $\text{Na} < \text{B}$

62. Choose the correct matching

	Column-I	Column-II
(1)	$[\text{Xe}] 4f^{14}5d^76s^2$	Ure
(2)	$5s^25p^2$	Ba
(3)	$[\text{Kr}] 4d^{10}5s^2$	Cd
(4)	$[\text{Ne}] 3d^54s^1$	Cu

63. Select the incorrect statement

- (a) P-orbital can accommodate $6e^-$
 (b) $\text{Na} \xrightarrow{\text{IE}_1} \text{Na}^+ \xrightarrow{\text{IE}_2} \text{Na}^{+2} \xrightarrow{\text{IE}_3} \text{Na}^{+3}$;

The correct order of Ionisation Energy is $\text{IE}_3 > \text{IE}_1 > \text{IE}_2$

- (c) Gd is Inner transition element.
 (d) B is non metal.

- (1) Only a (2) a and b
 (3) b and c (4) a, b and d

64. Which of the following ion has maximum Electron affinity -

- (1) O^+ (2) S^+ (3) Cl^+ (4) F^+

65. How many total elements from the following are not transition elements.

Zr, Co, Ge, Cd, Hg, Au, Cs

- (1) 5 (2) 4 (3) 6 (4) 3

66. Which of the following ion has the highest value of Ionic Radius -

- (1) O^{2-} (2) B^{3+}
 (3) Li^+ (4) Mg^{+2}

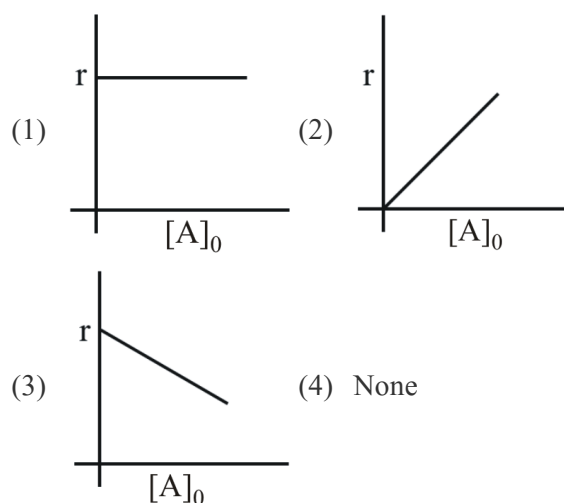
67. Which of following atom has maximum difference of Ionisation energy-

- (1) B, Tl (2) Al, Ga
 (3) B, In (4) Ga, In

68. Select the correct order of Atom/Ionic Radius is -

- (1) $\text{Zr}^{+4} \simeq \text{Hf}^{+4}$ (2) $\text{Eu}^{+3} > \text{Gd}^{+3}$
 (3) $\text{Eu} > \text{Gd}$ (4) All of these

69. For 1st order reaction, correct graph between rate and concentration of reactant will be :



70. **Assertion :** For a first order reaction $t_{1/2}$ is independent of the initial concentration of reactants.

Reason : For a first order reaction $t_{1/2}$ is twice the $t_{3/4}$.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.

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71. At 300 K the half-life of a sample of a gaseous compound initially at 1 atm is 100 s. When the pressure is 0.5 atm the half-life is 50 s. The order of reaction of compound is :-
 (1) 0 (2) 1 (3) 2 (4) 3
72. For reaction $A \rightarrow B$ rate constant is $1.2 \times 10^{-2} \text{ Ms}^{-1}$. What is concentration of B after 10 minute. If we start with 10 M of A :-
 (1) 2.8 M (2) 7.2 M (3) 8.2 M (4) 2.7 M
73. If $\frac{d[\text{NH}_3]}{dt} = 34 \text{ g hr}^{-1}$ for the reaction
 $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
 Then $-\frac{d[\text{H}_2]}{dt}$ is g h^{-1} .
 (1) 6 (2) 4 (3) 51 (4) 12
74. For a I^{st} order reaction what will be the time required to reach concentration from 1 M to 0.25 M if $t_{1/2}$ is 138 s :-
 (1) 276 s (2) 138 s
 (3) 69 s (4) 27.6 s
75. For a reaction $A + B \rightarrow C + 2D$, experimental results were collected for three trials and the data obtained are given below
- | Trial | [A], M | [B], M | Initial Rate, M s^{-1} |
|-------|--------|--------|---------------------------------|
| (1) | 0.40 | 0.20 | 55×10^{-4} |
| (2) | 0.80 | 0.20 | 5.5×10^{-3} |
| (3) | 0.80 | 0.40 | 2.2×10^{-2} |
- The correct rate law of the reaction is :
 (1) $\text{Rate} = k[\text{A}]^0 [\text{B}]^2$ (2) $\text{Rate} = k[\text{A}][\text{B}]^2$
 (3) $\text{Rate} = k[\text{A}][\text{B}]$ (4) $\text{Rate} = k[\text{A}][\text{B}]^0$
76. In a reaction $2A \rightarrow \text{Products}$, the Concentration of A decreases from 0.8 mol L^{-1} to 0.3 mol L^{-1} in 5 minutes then what is the average rate during this interval ?
 (1) $2.5 \times 10^{-2} \text{ M.min}^{-1}$
 (2) $10^{-1} \text{ M.min}^{-1}$
 (3) $1.25 \times 10^{-2} \text{ M.min}^{-1}$
 (4) $5 \times 10^{-2} \text{ M.min}^{-1}$
77. **Statement-I** : The rate of a first order reaction decreases with passage of time as the concentration of reactant decreases.
Statement-II : Value of rate constant decreases with decrease in concentration of reactant.
 (1) Statement-I and Statement-II both are true.
 (2) Statement-I is false and Statement-II is true.
 (3) Statement-I is true and Statement-II is false.
 (4) Statement-I and Statement-II both are false.
78. If rate is constant with respect to time then order will be :-
 (1) 1 (2) 0 (3) 2 (4) 3
79. Which of the following statement is incorrect –
 (1) In complex reaction molecularity of the slowest step is same as the order of the overall reaction if intermediate is not present.
 (2) The decomposition of gaseous ammonia on a hot platinum surface is a zero order reaction at low pressure.
 (3) The fraction of molecules having energy equal to or lower than threshold value are known as activated molecules.
 (4) Both 2 & 3 are incorrect.

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80. Fraction of molecules of reactants having energy equal to or greater than activation energy is :-

- (1) $e^{-\frac{E_a}{RT}}$ (2) $Ae^{-\frac{E_a}{RT}}$
 (3) $-\frac{E_a}{RT}$ (4) $\frac{E_a}{RT}$

81. The rate of a reaction quadruples when the temperature changes from 290 K to 310 K. Find the energy of activation of the reaction assuming that it does not change with temperature ?

- (1) $\frac{2.303R \times 310 \times 290}{20}$
 (2) $\frac{0.693R \times 310 \times 290}{20}$
 (3) $\frac{1.38R \times 310 \times 290}{20}$
 (4) $\frac{20}{1.38R \times 310 \times 290}$

82. A first order reaction takes 60 min. for 99.9% decomposition then half life is :

- (1) 60 min (2) 3 min (3) 2 min (4) 6 min

83. Match the terms of column-I with column-II

Column-I		Column-II	
A	Hydrolysis of ester in acidic medium	p	Zero order reaction
B	Radioactive decay	q	Pseudo first order reaction
C	Thermal decomposition of HI on gold surface	r	First order reaction

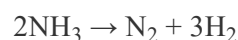
- (1) $A \rightarrow p, B \rightarrow q, C \rightarrow r$
 (2) $A \rightarrow q, B \rightarrow r, C \rightarrow p$
 (3) $A \rightarrow r, B \rightarrow q, C \rightarrow p$
 (4) $A \rightarrow q, B \rightarrow p, C \rightarrow r$

84. Half life of a radioactive substance is 10 years. What fraction of the substance remains after 20 years.

- (1) $\frac{1}{2}$
 (2) $\frac{1}{4}$
 (3) $\frac{1}{8}$
 (4) $\frac{1}{16}$

85. The decomposition of ammonia on platinum surface is zero order reaction what is the rate of production of H_2 .

If $K = 5 \times 10^{-4} \text{ mol L}^{-1} \text{ S}^{-1}$?



- (1) $3 \times 10^{-4} \text{ mol L S}^{-1}$
 (2) $5 \times 10^{-4} \text{ mol L S}^{-1}$
 (3) $1.5 \times 10^{-3} \text{ mol}^{-1} \text{ L S}^{-1}$
 (4) $1.5 \times 10^{-3} \text{ mol L}^{-1} \text{ S}^{-1}$

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

86. The incorrect option or statement is :-

- (1) The equation of 2nd IP is $X(g) \rightarrow X^{+2} + 2e^{-}$
 (2) 2nd IP of an element is lower than its 3rd IP
 (3) $\text{Cl}^{-} > \text{F}^{-}$ (I.E)
 (4) Ionisation enthalpies are always positive for a neutral atom.

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87. The correct match of contents in Column-I with those in Column-II is :

	Column-I		Column-II
(A)	He	(i)	High electron gain enthalpy
(B)	Cl	(ii)	Electropositive element
(C)	Ca	(iii)	IA group
(D)	Li	(iv)	Highest ionisation energy

- (1) A-(iii), B-(i), C-(ii), D-(iv)
 (2) A-(iv), B-(iii), C-(ii), D-(i)
 (3) A-(ii), B-(iv), C-(i), D-(iii)
 (4) A-(iv), B-(i), C-(ii), D-(iii)

88. **Assertion (A):** The ionic radii of O^{2-} and Mg^{2+} are same.

Reason (R): Both O^{2-} and Mg^{2+} are isoelectronic species.

In the light of the above statements, choose the correct answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (3) (A) is true but (R) is false.
 (4) (A) is false but (R) is true.

89. Which of the following process is an endothermic process

- (1) $O(g) + e^- \rightarrow O^-(g)$
 (2) $Cl(g) + e^- \rightarrow Cl^-(g)$
 (3) $N(g) + e^- \rightarrow N^-(g)$
 (4) $S(g) + e^- \rightarrow S^-(g)$

90. Read the following statements -

- (a) II^{nd} I.E. will be always greater than I^{st} I.E.
 (b) Size order $Mg > Al > Mg^{+2} > Al^{+3}$
 (c) Group 17 elements have very less negative electron gain Enthalpy.
 (d) In two compound OF_2 and Na_2O the correct order of Electronegativity is $F > O > Na$
 How many total statements are correct -

- (1) 2 (2) 3 (3) 4 (4) 1

91. Select the incorrect order of Electron gain enthalpy (EA)

- (1) $P < S$ (2) $N < O$ (3) $N^+ < N$ (4) $O^- < O$

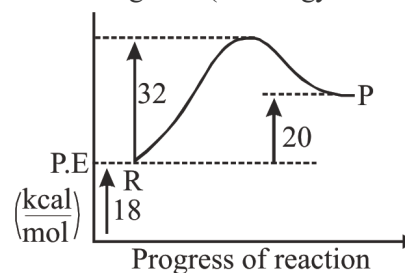
92. Which of the following is the correct order according to their given property -

- (1) $O^{2-} > Br^- > Se^{2-}$ Order of Radius
 (2) $C < N > O < F$ Order of Ionisation Energy
 (3) $Li < B$ Order of Electron affinity.
 (4) $NaF > CsF$ Ionic Nature.

93. 50% of first order reaction completed in 16 min then 75% of reaction completed in

- (1) 30 min (2) 32 min (3) 64 min (4) 16 min

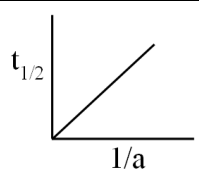
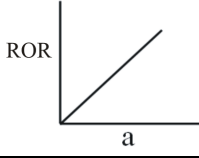
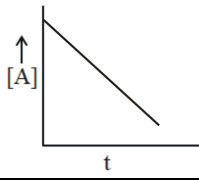
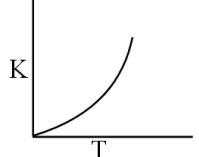
94. What is the threshold energy of reaction ($R \rightarrow P$) in represented diagram? (all energy in kcal/mol) :-



- (1) 12 kcal/mol (2) 14 kcal/mol
 (3) 50 kcal/mol (4) 70 kcal/mol

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95. Match the following :-

Column-I		Column-II	
(i)		(P)	Zero order
(ii)		(Q)	First order
(iii)		(R)	Second order
(iv)		(S)	$K = Ae^{-\frac{E_a}{RT}}$

- (1) (i)→P, (ii)→Q, (iii)→R, (iv)→S
 (2) (i)→S, (ii)→R, (iii)→Q, (iv)→P
 (3) (i)→R, (ii)→Q, (iii)→P, (iv)→S
 (4) (i)→Q, (ii)→P, (iii)→S, (iv)→R

96. A gaseous reaction $A_{2(g)} \longrightarrow B_{(g)} + \frac{1}{2}C_{(g)}$
 Shows increase in pressure from 100 mm of Hg to 120 mm of Hg in 5 minute. What will be rate of disappearance of A_2 (in mm of Hg/min) :-

- (1) 8 (2) 4 (3) 40 (4) 10

97. The hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below :-

- (1) $A_2 \rightleftharpoons A + A$ (fast)
 (2) $A + B_2 \rightarrow AB + B$(slow)
 (3) $A+B \rightarrow AB$ ----- (fast)

The order of the reaction is :-

- (1) 1.5 (2) 2 (3) Zero (4) 1

98. The factors affecting the rate constant of reaction are –

- I. Temperature
 II. Pressure
 III. Concentration of reactant or product
 IV. Catalyst

Choose the correct option from the alternative given below.

- (1) I, II and III
 (2) I, III and IV
 (3) I, IV
 (4) All of these

99. In a zero order reaction if the temperature is increased from 10°C to 40°C . The rate of reaction will become :-

- (1) 512 times (2) 8 times
 (3) 256 times (4) 64 times

100. Consider a first order gas phase decomposition reaction given below :



The initial pressure of the system before decomposition of A was P_i After lapse of time 't' total pressure is P_t .

The rate constant of reaction is :-

- (1) $K = \frac{2.303}{t} \log \frac{P_i}{P_i - x}$
 (2) $K = \frac{2.303}{t} \log \frac{P_i}{2P_i - P_t}$
 (3) $K = \frac{2.303}{t} \log \frac{P_i}{2P_i + P_t}$
 (4) $K = \frac{2.303}{t} \log \frac{P_i}{P_i + x}$

Topic : HUMAN REPRODUCTION (INTRODUCTION TO FERTILISATION), SEXUAL REPRODUCTION IN FLOWERING PLANTS**SECTION-A****Attempt All 35 questions**

101. In the given list how many structures are included in male reproductive system ?
Testes, Accessory ducts, Accessory glands, External genitalia.

- (1) 3 (2) 4
(3) 2 (4) 1

102. Which of the following statement is false ?

- (1) Humans are sexually reproducing and viviparous.
(2) Implantation is followed by formation of blastocyst.
(3) Sperm formation continues even in old man.
(4) Gestation is followed by parturition.

103. Secretions of which of the following glands helps in lubrication of penis ?

- (1) Prostate gland (2) Bulbourethral gland
(3) Both (1) and (2) (4) Seminal vesicle

104. Which of the following is not a part of female accessory ducts ?

- (1) Vagina (2) Mons pubis
(3) Oviduct (4) Uterus

105. The last part of oviduct which has a narrow lumen and joins the uterus is ?

- (1) Ampulla (2) Infundibulum
(3) Isthmus (4) Fimbriae

106. Which of the following pair is not matched correctly ?

- (1) Length of fallopian tube $\Rightarrow$ 10-12 cm
(2) Number of testicular lobules in a male $\Rightarrow$ 500
(3) Width of testis $\Rightarrow$ 4-5 cm
(4) Length of ovary $\Rightarrow$ 2-4 cm

107. Which layer of uterus undergoes cyclical changes during menstrual cycle ?

- (1) Perimetrium (2) Tunica vasculosa
(3) Myometrium (4) Endometrium

108. Tiny finger like structure which lies at the upper junction of two labia minora is ?

- (1) Mons pubis (2) Hymen
(3) Labia majora (4) Clitoris

109. **Statement-I** : Formation of ovum ceases in women around the age of fifty years.

Statement-II : Each testicular lobule contains 1-3 highly coiled seminiferous tubules in which sperm are produced.

Choose the correct answer from the given options :-

- (1) Both statement-I and II are correct
(2) Both statement-I and II are incorrect
(3) Statement-I is correct but statement II is incorrect
(4) Statement-I is incorrect but statement II is correct

110. Nurse cells are also termed as ?

- (1) Sertoli cells (2) Leydig cells
(3) Interstitial cells (4) Spermatogonia

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111. In the given list how many structures are present inside the testes ?
Immunologically competent cell, Leydig cells, Sertoli cells, Spermatogonia.
(1) 2 (2) 4 (3) 1 (4) 3
112. Which of the following hormone is released in case of excess spermatogenesis ?
(1) ABP (2) Testosterone
(3) FSH (4) Inhibin
113. Which statement is false for sperm ?
(1) A plasma membrane envelops only the head of sperm.
(2) The sperm head containing a haploid elongated nucleus.
(3) The middle piece contains numerous mitochondria.
(4) It is a microscopic structure
114. Which of the following is characterised by a fluid filled cavity called antrum ?
(1) Primordial follicle
(2) Secondary follicle
(3) Tertiary follicle
(4) Primary follicle
115. Which of the following is not another name of luteal phase of menstrual cycle ?
(1) Secretory phase
(2) Proliferative phase
(3) Progesteronic
(4) Postovulatory phase
116. Which of the following pair is not matched correctly ?
(1) Primary follicles in a female at puberty $\Rightarrow$ 1,20,000 - 1,60,000
(2) First menstruation at puberty $\Rightarrow$ Menarche
(3) Meiotic first division of oogenesis $\Rightarrow$ Equal division
(4) Arrest of meiosis-I during oogenesis $\Rightarrow$ Prophase-I
117. In a female if there is no entry of sperm then how many structures in the given list will not be formed ?
1st polar body, 2nd polar body, Secondary oocyte, Primary oocyte.
(1) 2 (2) 1 (3) 4 (4) 3
118. Read the following statement (A-D) :-
(A) Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.
(B) Secondary oocyte retains the bulk of the nutrient rich cytoplasm of the primary oocyte.
(C) In testis, the immature germ cells (spermatogonia) produce sperms by spermatogenesis that begins at puberty.
(D) Primary oocyte is surrounded by a layer of granulosa cells called the primary follicles.
Which of the above statements are true :-
(1) A, B, C and D (2) B only
(3) B,C and D only (4) A and C only
119. In oogenesis, the 2nd polar body is formed inside which structure ?
(1) Ovary (2) Uterus
(3) Peritoneal cavity (4) Fallopian tube

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120. Fusion of pronuclei of male and female gametes is ?
 (1) Amphimixis (2) Plasmogamy
 (3) Karyogamy (4) Ovulation
121. Which of the following hormone leads to increase in antral fluid in the ovarian follicles ?
 (1) ABP (2) LH
 (3) Progesterone (4) Inhibin
122. Which of the following hormone increases the secretory nature of endometrium ?
 (1) FSH (2) LH
 (3) Progesterone (4) Estrogen
123. Which of the following statement is true ?
 (1) Each seminiferous tubule is lined on the out side by germinal epithelium.
 (2) The hymen can never be torn by a sudden fall or jolt or insertion of vaginal tampon.
 (3) In humans, maturation phase is the longest phase of oogenesis.
 (4) On the 23rd day of menstrual cycle the thickness of uterine wall is minimum
124. **Assertion (A)** : Breast feeding during the initial period of infant growth is recommended by doctors.
Reason (R) : Colostrum contains antibodies (mainly IgA) absolutely essential to develop resistance for the new born babies.
 Choose the correct answer from the given options :-
 (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
 (2) (A) is correct but (R) is not correct.
 (3) (A) is not correct but (R) is correct.
 (4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

125. **Assertion (A)** : The presence of hymen is not a reliable indicator of virginity.
Reason (R) : In some women hymen persists even after coitus.
 Choose the correct answer from the given options :-
 (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
 (2) (A) is correct but (R) is not correct.
 (3) (A) is not correct but (R) is correct.
 (4) Both (A) and (R) correct and (R) is the correct explanation of (A).

126. The regions outside the seminiferous tubules called interstitial spaces contains :-
 (1) Small blood vessels
 (2) Interstitial cells
 (3) Immunologically competent cells
 (4) All of the above

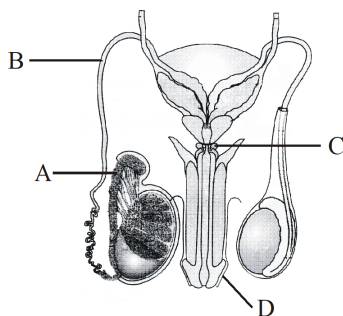
127. Select the correct match from following table.

(A)	ICSH	(i)	Released by hypothalamus
(B)	FSH	(ii)	Stimulate leydig cells
(C)	ABP	(iii)	Stimulate Sertoli cells
(D)	GnRH	(iv)	Released by sertoli cells

- (1) A – ii ; B – iii ; C – iv ; D – i
 (2) A – iii ; B – ii ; C – i ; D – iv
 (3) A – i ; B – ii ; C – iii ; D – iv
 (4) A – iv ; B – iii ; C – ii ; D – i

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128. It is a diagrammatic view of male reproductive system. Choose the correct option about A, B, C and D.



	A	B	C	D
(1)	Vasa efferentia	Vas deferens	Prostate	Foreskin
(2)	Epididymis	Vasa efferentia	Seminal Vesicle	Glans penis
(3)	Rete testis	Ureter	Bulbo-Urethral gland	Glans penis
(4)	Epididymis	Vas deferens	Bulbo-Urethral gland	Foreskin

129. 50 secondary oocytes in female and 50 secondary spermatocytes in male give rise to :-

- (1) 100 ova and 100 sperms
- (2) 200 ova and 50 sperms
- (3) 100 ova and 200 sperms
- (4) 50 ova and 100 sperms

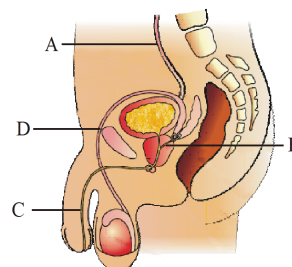
130. Capacitation refers to changes in the :-

- (1) Ovum before fertilization.
- (2) Ovum after fertilization.
- (3) Sperm after fertilization.
- (4) Sperm before fertilization.

131. The correct sequence of spermatogenetic stage leading to the formation of sperm in a mature human testis is :-

- (1) Spematogonia - Spermatids - Spermatocytes - Sperms
- (2) Spermatocyte - Spematogonia - Spermatids - Sperms
- (3) Spematogonia -Spermatocytes - Spermatids - Sperms
- (4) Spermatids - Spermatocytes - Spematogonia - Sperms

132. It is a diagrammatic sectional view of male reproductive system, In which identify common path for the semen and urine :

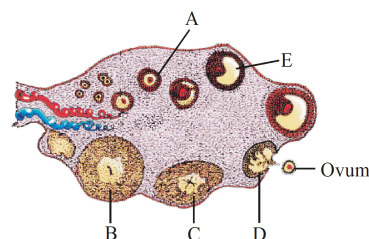


- (1) A (2) B (3) C (4) D

133. Seminal plasma in human is rich in :

- (1) Glucose, fructose, Ribose
- (2) Fructose, Calcium, Certain enzymes
- (3) Ribose, Testosterone, Mucus
- (4) Testosterone, ABP and Sperms

134. Observe the following diagram and answer :



Which structure secrete progesterone :-

- (1) A and E
- (2) E and D
- (3) B and E
- (4) B and C

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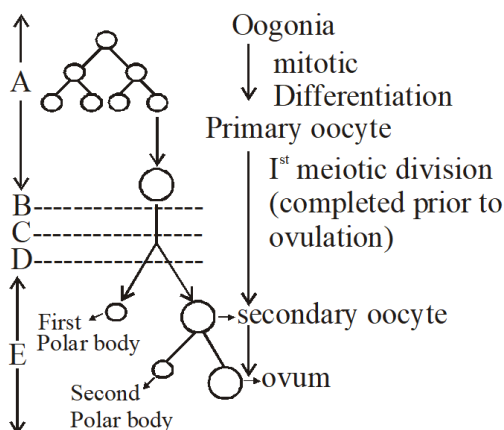
135. Which statement is incorrect ?

- (1) The stroma of ovary is divided into two zones, cortex and medulla.
- (2) In the later phase of pregnancy, a hormone called relaxin is also secreted by the ovary.
- (3) Corona radiata is a primary egg membrane.
- (4) Hyaluronic acid is found between the cells of corona radiata.

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

136. The following refers to schematic representation of oogenesis. Identify A to E correctly



- (1) (A) Fetal life (B) Birth (C) Puberty (D) Adult reproductive life (E) child hood
- (2) (A) Fetal life (B) Birth (C) Childhood (D) Puberty (E) Adult reproductive life
- (3) (A) Adult reproductive life (B) Birth (C) puberty (D) child hood (E) Fetal life
- (4) (A) Birth (B) child hood (C) Fetal life (D) puberty (E) adult reproductive life

137. Which statement is not correct for both testes and ovary :

- (1) Both are primary sex organs.
- (2) Both are developed from mesoderm.
- (3) Both are developed in abdominal cavity in Intra uterine life.
- (4) At the time of birth both descend down into extra abdominal space.

138. The _____ are the first haploid cells during the process of spermatogenesis.

- (1) Spermatogonia
- (2) Primary spermatocyte
- (3) Secondary spermatocyte
- (4) Spermatids

139. Choose the correct option for filling up the blanks:-

The human male ejaculates about _____ million sperms during a coitus of which, for normal fertility, at least _____ percent sperms must have normal shape and size and least _____ percent of them must show vigorous motility.

- (1) 100–200, 40, 60
- (2) 200–300, 60, 40
- (3) 300–400, 50, 30
- (4) 500, 70, 70

140. Choose the correct order for the path of sperm from the testes to outside the body.

- (1) vas deferens - epididymis - ejaculatory duct - penis
- (2) epididymis - vas deferens - ejaculatory duct - penis
- (3) ejaculatory duct - vas deferens - epididymis - penis
- (4) penis - ejaculatory duct - epididymis - vas deferens

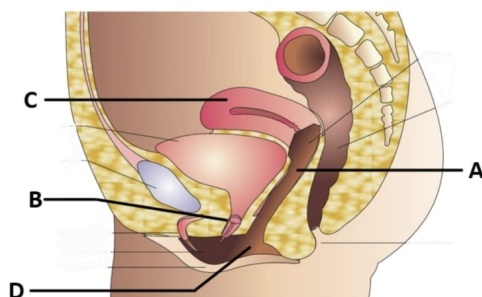
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141. Arrange following events of female reproductive cycle in (human) is sequence :-

- (A) Secretion of FSH
- (B) Growth of corpus luteum
- (C) Growth of follicles
- (D) Ovulation
- (E) Sudden increase in level of LH

- (1) C → A → D → B → E
- (2) A → C → E → D → B
- (3) A → D → C → E → B
- (4) B → A → C → D → E

142. Given figure is a diagrammatic sectional view of female pelvis showing reproductive system identify A,B,C and D :-



	A	B	C	D
(1)	Vagina	Urethra	Uterus	Vaginal orifice
(2)	Vaginal orifice	Cervix	Urinary bladder	Labia majora
(3)	Cervix	Urethra	Uterus	Vagina
(4)	Urethra	Vagina	Urinary bladder	Urethral orifice

143. Vasa efferentia are the ductules those connect :-

- (1) testicular lobules to rete testes
- (2) rete testes to epididymis
- (3) vas deferens to epididymis
- (4) epididymis to urethra

144. Correct sequence for milk ejection is :

- (1) Mammary tubule – Mammary duct – Lactiferous duct – Ampulla – Nipple
- (2) Mammary duct – Mammary tubule – Ampulla – Lactiferous duct – Nipple
- (3) Mammary tubule – Mammary duct – Ampulla – Lactiferous duct – Nipple
- (4) Lactiferous duct – Mammary tubule – Mammary duct – Ampulla – Nipple

145. Choose the **incorrect** statement :-

- (1) Oogonia are formed and added after birth
- (2) Oogenesis is initiated during the embryonic development
- (3) No more mother gamete cells are formed in females after birth
- (4) 60000–80000 Primary follicle are present in each ovary at puberty

146. How many of the following structures in the given list contain 46 chromosomes ?

Spermatozoa, secondary oocyte, spermatogonia, secondary spermatocyte, primary spermatocyte, spermatid, polar body, primary oocyte, oogonia

- (1) Six
- (2) Four
- (3) Five
- (4) Seven

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147. Read the following statements carefully.

The(A)..... present on the inside wall of(B)..... multiply by(C)..... division. A(D)..... completes the(E)..... division leading to formation of two equal, haploid cells called(F)..... .

Which one of the following options gives the correct filling of A, B, C, D, E and F?

	(1)	(2)	(3)	(4)
(A)	Spermatogonia	Primary Spermatocyte	Spermatogonia	Primary Spermatocyte
(B)	Seminiferous tubule	Sertoli Cell	Sertoli Cell	Seminiferous tubule
(C)	Mitotic	Mitotic	First meiotic	First meiotic
(D)	Primary Spermatocyte	Secondary Spermatocyte	Secondary Spermatocyte	Spermatid
(E)	First meiotic	First meiotic	Second meiotic	Second meiotic
(F)	Secondary Spermatocyte	Spermatid	Spermatid	Sperm

148. Match the following and choose the correct answer :-

(A)	Spermatogenesis	(i)	Vagina
(B)	Capacitation	(ii)	Ovary
(C)	Folliculogenesis	(iii)	Testis
(D)	Fertilisation	(iv)	Fallopian tube

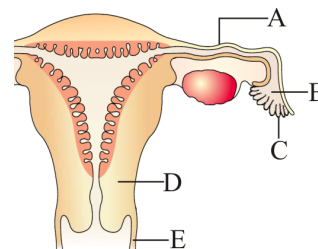
(1) A-(iv), B-(i), C-(iii), D-(ii)

(2) A-(iii), B-(ii), C-(iv), D-(i)

(3) A-(iii), B-(i), C-(ii), D-(iv)

(4) A-(iii), B-(iv), C-(ii), D-(i)

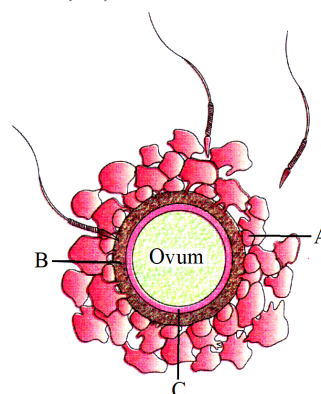
149.



See the diagram and Identify A,B,C,D,E :-

	A	B	C	D	E
(1)	Infundibulum	Ovary	Fimbriae	Cervix	Vagina
(2)	Fallopian tube	Infundibulum	Cervix	Vagina	Isthmus
(3)	Isthmus	Infundibulum	Fimbriae	Cervix	Vagina
(4)	Cervix	Ovary	Vagina	Fimbriae	Isthmus

150. It is a diagrammatic presentation of ovum which is surrounded by few sperms, choose the correct option about A, B, and C :-



	A	B	C
(1)	Vitelline membrane	Zona Pellucida	Cells of the corona radiata
(2)	Zona Pellucida	Vitelline membrane	Cells of the corona radiata
(3)	Plasma membrane	Vitelline membrane	Perivitelline space
(4)	Cells of the corona radiata	Zona Pellucida	Perivitelline space

Topic : HUMAN REPRODUCTION (INTRODUCTION TO FERTILISATION), SEXUAL REPRODUCTION IN FLOWERING PLANTS**SECTION-A****Attempt All 35 questions****151.** Polycarpic plants are :-

- (1) Annual (2) Biennial
(3) Perennial (4) Fruiting once

152. Pollen grains :-

- (1) Are generally tetragonal
(2) Measuring 25-50 micrometers in diameter
(3) Has prominent three layered wall
(4) Formed inside megasporangium

153. Functional megaspore develops into polygonum type embryo sac by

- (1) One meiosis and two successive mitosis
(2) Two meiosis and two successive mitosis
(3) One meiosis and three successive mitosis
(4) Three successive mitosis

154. Select the **incorrect** pair :-

- (1) Embryo sac – Microgametophyte
(2) Pollen grain – Microspore
(3) Stamen – Microsporophyll
(4) Ovule – Megasporangium

155. One of the most resistant organic material present in the exine of pollen grain is :-

- (1) Pectocellulose (2) Sporopollenin
(3) Suberin (4) Cellulose

156. Pollen tablets are used as _____

- (1) As antibiotic (2) As antiseptic
(3) As food supplement (4) Anti allergen

157. In flowering plants, two male gametes are formed by mitotic division in

- (1) Generative cell (2) Central cell
(3) Synergid cell (4) Tube cell

158. Which of the following is not a feature of typical angiospermic anther ?

- (1) It consist of four microsporangia
(2) It is situated on distal end of filament
(3) It is site for microspore formation
(4) At maturity, when it dehisce, then it has two pollen sacs in each lobe

159. Tapetum cells of anther are usually :-

- (1) Diploid, uninucleate
(2) Haploid, binucleate
(3) Haploid, tetranucleate
(4) Polyploid, multinucleate

160. Typical anther of flowering plant is :

- (1) Ditheous and tetrasporangiate
(2) Ditheous and bisporangiate
(3) Monotheous and bisporangiate
(4) Monotheous and monosporangiate

161. Residual, persistent nucellus is :-

- (1) Perisperm (2) Endosperm
(3) Embryo sac (4) Aril

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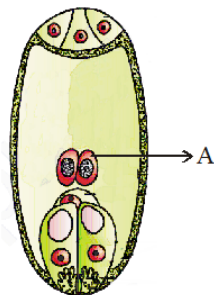
- 162.** The egg apparatus in an embryo sac consists of
- (1) Antipodal + egg cell
 - (2) Synergids + egg cell
 - (3) Synergids + antipodals + egg cell
 - (4) Synergids + polar nuclei + egg cell
- 163.** How many meiotic divisions are required to form 200 mature male gametophytes in angiosperm from microspore mother cell ?
- (1) 50 (2) 25 (3) 100 (4) 200
- 164.** First cell of female gametophyte is :-
- (1) Functional megaspore
 - (2) Microspore mother cell
 - (3) Megaspore mother cell
 - (4) Egg cell
- 165.** Given below are two statements :-
Statement-I : Geitonogamy is functionally cross-pollination involving a pollinating agent.
Statement-II : Geitonogamy is genetically similar to autogamy since pollen grains come from the same plant.
 Choose correct answer from the options given below :-
- (1) Both statement-I and statement-II are incorrect.
 - (2) Statement-I is correct but statement-II is incorrect.
 - (3) Both statement-I and statement-II are correct.
 - (4) Statement-I is incorrect but statement-II is correct.
- 166.** In which one of the following, both autogamy and geitonogamy are prevented ?
- (1) Wheat (2) Papaya
 - (3) Castor (4) Maize
- 167.** What is true about cleistogamy ?
- (1) It occurs in unisexual flowers
 - (2) It produces assured seed-set even in the absence of pollinators
 - (3) It leads to introduction of new useful characters
 - (4) It is a method of geitonogamy
- 168.** Which one of the following statements is not true ?
- (1) Micropyle facilitates entry of pollen tube into seed, during germination.
 - (2) Pollen grains are rich in nutrients, and they are used in the form of tablets and syrups
 - (3) Pollen grains of some plants cause severe allergies and bronchial afflictions in some people
 - (4) The flowers pollinated by flies and bats secrete foul odour to attract them
- 169.** Maize and grasses show :-
- (1) Entomophily (2) Hydrophily
 - (3) Chiropterophily (4) Anemophily
- 170.** Which of the following is not an adaptation for anemophily ?
- (1) Presence of well exposed stamens
 - (2) Often presence of single ovule in an ovary
 - (3) Sticky pollen grain
 - (4) Large and often feathery stigma
- 171.** Features like large feathery stigma, non-sticky pollen grains and single ovule in each ovary are characteristics of plants pollinated by :-
- (1) Insect (2) Water
 - (3) Air (4) Birds

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- 172.** When an octaploid male plant is crossed with octaploid female plant then ploidy of endosperm and embryo would be, respectively
- (1) $12n, 8n$
 - (2) $8n, 12n$
 - (3) $6n, 6n$
 - (4) $8n, 8n$
- 173.** Which one of the following statements is not true?
- (1) In the grass family cotyledon is called scutellum.
 - (2) Five nuclei and three male gametes participate in double fertilization
 - (3) Embryo develops at the micropylar end of embryo sac where the zygote is situated
 - (4) Orchid seeds are non-endospermic
- 174.** Read the following statements carefully :-
- (a) Pollen tube contains two male gametes
 - (b) One male gamete is motile and other is non motile
 - (c) Both male gametes are non-motile
 - (d) Pollen tube shows chemotropic movement
- How many statements are correct about flowering plants ?
- (1) 1
 - (2) 2
 - (3) 3
 - (4) 4
- 175.** Pollen tube releases the two male gametes into the :
- (1) One male gamete in cytoplasm of synergids and other in cytoplasm of central cell
 - (2) Both in cytoplasm of central cell
 - (3) Cytoplasm of antipodal cell
 - (4) Both in cytoplasm of synergids
- 176.** Select correct for embryo development in flowering plants
- (1) Develops at chalazal end of ovule
 - (2) Early stages of embryo development differs in monocots and dicots
 - (3) Most zygote divide only after certain amount of endosperm is formed
 - (4) Zygotes divide by meiosis division
- 177.** Select the event from the following which is not referred as pollen pistil interaction :-
- (1) Deposition of pollen on the stigma
 - (2) Entry of pollen tube into ovule
 - (3) Growth of pollen tube into the style
 - (4) Division of microspore mother cell in anther
- 178.** How many of the following statements are true?
- (A) Embryo development precedes endosperm development in angiosperm
 - (B) White kernel of coconut is free nuclear endosperm
 - (C) *Vallisneria* is a dioecious plant and flowers are unisexual.
 - (D) Formation of megaspore from megaspore mother cell is called megasporogenesis.
- (1) Three
 - (2) Four
 - (3) Two
 - (4) One

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179. In the given figure, which of the following statements are true about the labelled region "A"



- (a) Participate in syngamy
(b) Participate in triple fusion
(c) Participate in embryo formation
(d) Participate in endosperm formation
- (1) a only (2) b and d only
(3) a, b and d only (4) a and b only
180. Identify the wrong statements regarding post fertilization development :-
- (1) The ovary wall develops into pericarp
(2) The ovary wall develops into seed coat
(3) The primary endosperm cell develops into endosperm
(4) The ovule develops into seed
181. **Assertion :-** Syngamy leads to formation of a specialised cell called zygote.
Reason :- One male gamete fertilise the egg cell in syngamy.
- (1) Both assertion and reason are true and reason is correct explanation of assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

182. Arrange the following terms in the correct development sequence :-

(A) Male gametes
(B) Pollen grain
(C) Sporogenous tissue
(D) Pollen mother cell
(E) Microspore tetrad

- (1) C, D, E, B, A
(2) B, C, D, A, E
(3) C, D, A, B, E
(4) B, D, C, A, B

183. During an experiment, vegetative nucleus was destroyed inside the pollen tube. Immediately afterwards it was observed that the growth of pollen tube was stopped and pollen tube could not reach up to ovule, because-

- (1) Male gametes were not developed
(2) Growth of pollen tube is controlled by vegetative nucleus
(3) Vegetative nucleus secretes chemicals
(4) All of the above

184. Match the following

A.	Body of ovule fuses with funicle here	I.	Chalaza
B.	Basal part of the ovule	II.	Funicle
C.	Attached to placenta	III.	Nucellus
D.	Contains nutritive tissues	IV.	Hilum

- (1) A – III, B – IV, C – I, D – II
(2) A – IV, B – I, C – II, D – III
(3) A – II, B – IV, C – I, D – III
(4) A – III, B – IV, C – I, D – II

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185. Consider the following five statements (a-e) and select the option which includes all the correct ones only :-

- (a) Vegetative cell is bigger, has abundant food reserve and a large irregularly shaped nucleus.
- (b) Over 60 percent of angiosperms, pollengrains are shed at 3-celled stage.
- (c) Pollen grains can be stored for years in liquid nitrogen (-196°C).
- (d) Members of Rosaceae, Leguminosae and Solanaceae the pollen grains are viable for months.
- (e) Pollen grains are poor in nutrients.

- (1) a, b, c, d
- (2) a, c, d
- (3) b, c, d
- (4) a, c, d, e

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

186. A flower has 10 microspore mother cells then how many pollen grains will be produced in that flower ?

- (1) 50 (2) 10 (3) 20 (4) 40

187. In over 60 percent of Angiosperms, Pollen grains are shed at :

- (1) 2-celled stage
- (2) 3-celled stage
- (3) 4-celled stage
- (4) 1-celled stage

188. Aero allergens that came into India as a contaminant with imported wheat become ubiquitous in occurrence :-

- (1) *Chenopodium* (2) *Parthenium*
- (3) *Amaranthus* (4) Sorghum

189. Filiform apparatus is characteristic of :-

- (1) Synergids (2) Egg
- (3) Anther wall (4) Antipodals cells

190. In grass embryo, epicotyl has a shoot apex and a few leaf primordia enclosed in a hollow foliar structure called :-

- (1) Coleoptile (2) Coleorhiza
- (3) Scutellum (4) Aleurone layer

191. One ovule is present in the ovary of:-

- (1) Paddy (2) Papaya
- (3) Watermelon (4) Orchids

192. Both chasmogamous and cleistogamous flowers are present in :-

- (1) Sunflower (2) *Commelina*
- (3) Pea (4) *Capsella*

193. Which of the following is not correct for double fertilization?

- (1) Two types of fusion, syngamy and triple fusion take place
- (2) The synergid cell after syngamy becomes the primary endosperm cell
- (3) Double fertilization was discovered by Nawaschin
- (4) It is of universal occurrence among angiosperms

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194. Which of the following is a significance of triple fusion during double fertilisation in angiospermic plants?

- (1) Essential for the formation of zygote
- (2) Essential for the formation of endosperm
- (3) Essential for the formation of embryo
- (4) Essential for the formation of cotyledons in seed

195. The primary endosperm cell & zygote will be _____ & _____ in a seed :-

- (1) Embryo & Endosperm
- (2) Embryo & Integuments
- (3) Integuments & Embryo
- (4) Endosperm & Embryo

196. Which one of the following is not a post-fertilisation event?

- (1) Development of PEC into endosperm
- (2) Development of ovary into fruit
- (3) Maturation of ovule into seed
- (4) Development of 7 celled-8 nucleate embryo sac

197. There are different developmental stages of embryo in dicot :-

- (A) Globular embryo
- (B) Heart shaped
- (C) Zygote
- (D) Proembryo
- (E) Torpedo (mature embryo)

Arrange them in sequence

- (1) C, D, A, B, E (2) B, A, C, D, E
- (3) D, C, A, B, E (4) A, E, D, C, B

198. Match the following Column - I and II, select correct option indicating a correct match?

	Column-1 (Type of Pollination)		Column-II (Example)
(i)	Entomophily	(a)	<i>Zostera</i>
(ii)	Ephihydrophily	(b)	<i>Pinus</i>
(iii)	Hypohydrophily	(c)	<i>Amorphophallus</i>
(iv)	Anemophily	(d)	<i>Vallisneria</i>

- (1) (i)-(b); (ii)-(d); (iii)-(c); (iv)-(a)
- (2) (i)-(c); (ii)-(a); (iii)-(d); (iv)-(b)
- (3) (i)-(c); (ii)-(d); (iii)-(a); (iv)-(b)
- (4) (i)-(a); (ii)-(c); (iii)-(d); (iv)-(b)

199. Read the following statements and find out the incorrect statement.

- (a) Plants use two abiotic (wind and water) and one biotic (animals) agents to achieve pollination.
- (b) Majority of plants use abiotic agents for pollination.
- (c) Only a small proportion of plants uses biotic agents.
- (d) Pollination by water is common among abiotic pollinators.
- (e) Pollination by wind is quite rare in flowering plants and is restricted to about 30 genera mostly monocotyledons.

- (1) a, b, c and d (2) b, c, d and e
- (3) a, c, d and e (4) b and d only

200. How many of the following structure in angiosperm plant have haploid set of chromosomes in cell ?

Microspore, generative cell, synergid, nucellus, integument, antipodal cell, archesporial cell, endosperm, zygote, vegetative cell of pollen

- (1) Six (2) Eight (3) Five (4) Three

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1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

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